

Product Requirements Document (PRD)

KollegeApply Google Ads Intelligence & Operations Platform ("Command Center")

Status: Live (production on Render) - **Last updated:** 2026-08-13

Owner (outgoing): Deepak Mishra - **Audience:** product, engineering, ops leadership, and whoever maintains/extends this platform next.

> This PRD is paired with [docs/KNOWLEDGE_TRANSFER.md](#) (the technical KT). This document is the **what & why**; the KT is the **how**.

1. Vision & problem statement

KollegeApply runs Google Ads for ~70 colleges across multiple Google Ads accounts (under one MCC). The team (an Ads-Operations manager + Account Managers, "AMs") needs to:

- 1 **See the truth fast** — real spend, clicks, leads, CPL, CTR, CPM across every account/campaign without logging into Google Ads and switching accounts.
- 1 **Catch waste & problems** — disapproved ads, budget-limited campaigns, low-quality keywords, wasteful search terms, zero-conversion spend.
- 1 **Plan new campaigns with data** — generate ad copy, keywords, negatives, budgets and forecasts grounded in real history (or peer benchmarks for brand-new colleges).
- 1 **Enforce accountability** — every plan is reviewed/approved before launch; budgets are allocated and tracked; managers own their accounts.

The platform is a **read + intelligence layer on top of Google Ads** (it never writes to Google Ads — see §9). It syncs Google Ads data into PostgreSQL as historical snapshots and exposes dashboards, an AI ad-copy planner, an approval workflow, and budget governance.

Non-goal: it is not an ad-serving system and does not push changes to Google Ads. All "approve/launch" actions are **records/gates**, not API writes.

2. Users & roles

Authentication is **Google sign-in** (OpenID Connect), restricted to approved Workspace domains. Two roles:

Role	Who	Can see	Can do
Admin	Ads-Ops lead, leadership	All accounts	Everything: assign accounts to AMs, set overall account budgets, download exports, approve plans, send admin emails, manage users

Role	Who	Can see	Can do
Manager (AM)	Account managers	Only accounts assigned to them (server-enforced)	View their accounts; generate ad copy; plan weekly/monthly budgets for their accounts; submit plans for approval. Cannot download exports, set the overall account budget, or see other AMs' accounts

Scoping is **enforced on the backend** (an AM cannot fetch another account's data by tampering with the request), not just hidden in the UI. See KT §"Auth & RBAC".

How an AM gets access to an account: an admin assigns it in **Users & Access**, *or* the AM is set as the owner of a campaign in **Campaign Accountability** (ownership grants account access automatically).

3. Feature modules (requirements)

Each module below lists what it does and the key requirements it satisfies. Nav is grouped: **Performance, AI Planner, Accountability, Reports, Admin**.

3.1 Overview (*Executive Overview*)

- 1 Top KPI card: total spend for the selected date range + Clicks, Impressions, **CPM**, Conversions, Campaigns.
- 1 Latest-day tiles (spend/clicks/impressions), Avg CTR/CPC, active campaigns/keywords, limited-by-budget, disapproved ads, low-QS keywords.
- 1 Spend trend chart + open-alerts summary + top-priority queue.
- 1 **Account breakdown** — expandable table: spend per account for the selected window -> drill into each account's campaigns with **landing page link, impressions, clicks, CTR, CPC, CPM, conversions, CPL**. Campaign names are **clickable** -> **Campaign Detail**. Admins can filter by account; AMs see only theirs. **Download Excel** (admin only).
- 1 Date filter (top bar): presets (7d...All time) or a custom range; account filter.

3.2 Campaign Detail (*/campaigns/:id*)

Full per-campaign drill-down for the selected window: KPI tiles (spend/impr/clicks/CTR/CPC/CPM/conv/CPL), spend & clicks/CTR history charts, keyword table (with quality score + issues), search terms, and **suggested negative keywords** (zero-conversion spend). Bidding strategy & status are read from the campaign's own snapshots (reliable), not the dimension record.

3.3 Accounts, Campaign Health, Campaign Explorer, Keyword Health, Search Terms, Trends, Budgets

Standard analytical views over the synced data, each respecting the account + date filters and RBAC scope. Campaign Health scores campaigns; Keyword Health flags low-QS/low-CTR/wasteful keywords; Search Terms explores actual queries; Trends shows day-over-day series; Budgets shows Google-Ads budget utilization.

3.4 AI Ad Copy Generator (*/ai/ad-copy*) — *the flagship planner*

Type a **college name** + paste a **landing page URL** -> generate a complete, data-grounded plan:

- 1 **Responsive Search Ads** (headlines + descriptions) — deterministic engine by default; hybrid LLM (Anthropic or Gemini) when an API key is set.
- 1 **Keyword intelligence** — scored keywords with intent, match type, search volume, suggested bid. **Editable:** users can remove suggested keywords, add their own (search volume auto-fetched from **Google Keyword Planner**), and change intent/match. Edits are tagged and folded into the ad-group grouping. Match types: **Phrase / Exact / Both**.
- 1 **Ad groups** (paste-ready, per match type), **negative keywords**, **landing-page quality score**, **budget plan + forecast** (clicks, leads, CPL), **seasonality**, **CPL optimizer**, **reverse planner**, **setup guide**.
- 1 **Cold-start (new college, no history):** works from the landing page + brand; keywords synthesized/planner-fetched; forecast anchored to the **median CPC/CVR across existing colleges** (peer benchmark) instead of a flat constant. Ambiguous names (e.g. "Great Lakes") are education-qualified and off-topic keywords filtered out.
- 1 **Export** to Excel/CSV/JSON (admin only), reflecting keyword edits.

Requirement — provenance discipline: every number is labelled **REAL** (from Google), **ARITHMETIC** (computed), or **ESTIMATE/TARGET** (assumption, e.g. CVR). Never present an estimate as actual.

3.5 Landing Page Auditor (/landing-auditor)

Paste any URL -> audit tracking (GTM/GA4/Ads conversion/Pixel/consent/remarketing), content quality, **course detail** (not just course names), **broken links** and **external/leaking links** (own subdomains treated as internal), page load speed, mobile viewport, forms, privacy/terms, message match -> a **reuse / reuse-with-fixes / rebuild** verdict + a category-scored quality score. Works with **no campaign/history** and for **all roles**.

3.6 Approval & accountability workflow

- 1 A generated plan moves **draft** -> **submitted** -> **approved | rejected | changes-requested**. Approval is a **record/gate**, never a Google write.
- 1 On **submit**, a **KPI gate** requires budget + target leads + target CPL. The full plan is emailed to **all platform admins** (see §3.9) with **one-click Approve / Reject** links.
- 1 **Approve** -> the submitter is emailed "cleared to build in Google Ads". **Reject** -> the reviewer enters a reason; the submitter is emailed the reason to revise.
- 1 The email contains the strategy, ad copy, **keywords with search volumes** (added-by-user tagged, removed-by-user listed), landing score, negatives — so the reviewer can decide without opening the app.
- 1 **Campaign Accountability** page: per-campaign plan-vs-actual + per-ad-manager rollups; admins set the **overall account budget** here (see §3.8).

3.7 Budget governance (two layers)

- 1 **Overall account budget** (admin, on Campaign Accountability) — the total allocated to each account; tracked vs all-time spend + remaining.
- 2 **Budget Planner** (/weekly-budgets, AMs) — AMs plan **weekly (Mon–Sun)** and **monthly** budgets for their assigned accounts against the admin's overall allocation; spent/remaining tracked.

3.8 Weekly budget email (Monday, to admins)

Automated Monday summary of the week just ended: per account — **manager, budget, spent, remaining, impressions, clicks, leads, CTR, CPM, CPL**, plus **week-on-week deltas** (spend/leads/CPL/CTR, green=improved), **this-month** budget vs spend, and **all-time total** allocated vs spent. Can also be sent on demand ("Email admins now").

3.9 Admin: Users & Access + Data audit

Admins manage users (role, active, assigned accounts) and see an **Account data audit** (each account's latest data date + duplicate-record flags) to spot stale/duplicated accounts.

3.10 Reports, Priority Queue, Alerts, Execution Audit

CSV/Excel reports (daily/weekly/monthly), a prioritized "what to fix first" queue, an alerts system (budget-limited, disapprovals, etc.), and an execution audit.

4. Key workflows (end to end)

A. Daily ops: Admin/AM opens Overview -> checks spend/alerts/priorities -> drills into an account -> a campaign -> keywords/search terms -> acts in Google Ads directly.

B. New campaign: AM opens AI Ad Copy Generator -> types the college + landing URL + budget -> generates -> edits keywords -> submits for approval -> admin approves via email -> AM builds it in Google Ads using the exported plan.

C. Budget cycle: Admin sets each account's overall budget (Accountability) -> AMs plan weekly/monthly budgets (Budget Planner) -> Monday email reports adherence + WoW improvement to admins.

5. Data model (conceptual)

Google Ads entities are synced as **dimensions** (Account, Campaign, AdGroup, Ad, Keyword, SearchTerm, Budget) + **append-only daily snapshots** (metrics per day). On top: **AdCopyGeneration** (+ approval events, scorecard snapshots), **User/UserAccount** (auth + access), **AccountWeeklyBudget**, **AccountBudget** (monthly/total), **AuditLog**, **SyncLog**. Full ERD in KT §"Data model" and [docs/er-diagram.md](#).

6. Non-functional requirements

- 1 **Single process:** FastAPI serves the API **and** the built React app on one port (:8000). No separate frontend server.
 - 1 **Data provenance:** REAL vs ARITHMETIC vs ESTIMATE labelling is a hard requirement.
 - 1 **Security:** RBAC enforced server-side; downloads admin-only; secrets only in **.env**/Render env, never committed; Google sign-in domain-restricted.
 - 1 **Read-only to Google:** the platform never writes to Google Ads.
 - 1 **Resilience:** sync is idempotent (re-runnable); email/LLM/Planner failures degrade gracefully (never break a request).
 - 1 **Auditability:** mutating requests are logged; approvals have an event log.
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7. Known limitations & data-integrity notes (important)

- 1 **Duplicate account/campaign records:** the DB has duplicated campaign (and some account) records from historical double-imports, causing occasional id/name misalignment (e.g. a campaign's dimension name not matching its snapshot metrics). Mitigated in the UI where possible; the **Account data audit** (Users & Access) surfaces duplicates. A source-level dedup is the proper fix (KT §"Known issues").

- 1 **Render is a read-only mirror** with the scheduler **off** — it does **not** sync live. Fresh data comes from the local source-of-truth then a re-mirror (KT §"Deployment").
 - 1 **Keyword Planner** needs Standard Google Ads API access (confirmed working).
 - 1 **Conversion tracking** is absent on many accounts -> "leads" show 0 there (shown honestly, not fabricated).
 - 1 **Performance Max / Demand Gen** campaigns have no keywords/search terms (expected).
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8. Secrets that must be rotated (handover)

The following were exposed in chat/history during development and **must be rotated** by the successor: the Google **OAuth client secret**, the **SESSION_SECRET**, the **Brevo/Resend API key**, the Render **Postgres password**, and the historically-leaked **Gemini key** (in `.env.example` git history). See KT §"Secrets & rotation".

9. Roadmap / not-yet-built

- 1 Source-level **dedup** of duplicate accounts/campaigns.
 - 1 **Automatic** Monday budget email on Render (currently on-demand; needs a Render Cron or enabling the scheduler there).
 - 1 **Meta/Bing** integration (future).
 - 1 **Match-type & strategy reports** (partially present in Reports).
 - 1 Optional: real **PageSpeed Insights** for landing load speed; **write-back** to Google Ads (explicitly out of scope today).
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*Companion doc: `docs/KNOWLEDGE_TRANSFER.md`. Also see `docs/architecture.md`, `docs/api.md`, `docs/deploy-render.md`, `docs/google-ads-setup.md`, `docs/ai_ad_copy/.*`